

Day 6 – Operating System (OS Basics)

1 GUI vs CLI + Basic Linux/Windows Commands

Aaj hum OS ke **fundamentals + important commands** ko deep aur exam/interview level par samjhenge 🔥

1 GUI – Graphical User Interface

Basic Definition

GUI ek aisa interface hai jisme user **icons, buttons, windows** aur **mouse** ka use karke system operate karta hai.

 Example:

- Windows Desktop
 - File Explorer
 - Settings App
-

Key Points

- ✓ Easy to use (Beginner friendly)
 - ✓ No coding required
 - ✓ Visual interaction hota hai
 - ✓ Slower than CLI (comparatively)
-

Real Life Example

File delete karna:

 Right click → Delete

2 CLI – Command Line Interface

Basic Definition

CLI me user **commands** type karke system control karta hai.

 Example:

- Linux Terminal
 - Windows CMD / PowerShell
-

 Key Points

Key Points

- ✓ Fast execution
- ✓ Powerful for hackers & admins
- ✓ Requires command knowledge
- ✓ Automation possible

Real Life Example

File delete:

```
rm file.txt
```

IP Configuration Commands

☐ Windows Command

1 ipconfig

📁 System ka IP address show karta hai

```
ipconfig
```

- ✓ IPv4 address
- ✓ Gateway
- ✓ Subnet mask

Linux Command

1 ifconfig

📁 IP address check (Kali Linux)

```
ifconfig
```

⚠ Note: New systems me `ip a` bhi use hota hai

Admin / Root Access

2 sudo su

📁 Root user banne ke liye (Linux)

```
sudo su
```

- ✓ Full system control
- ✓ Admin privileges

File & Directory Commands

3 ls

 Current directory ke files list karta hai

```
ls
```

- ✓ Files + folders show

4 cd (Change Directory)

```
cd Downloads
```

 Folder ke andar jane ke liye

Back jana:

```
cd ..
```

5 pwd

 Current location show karta hai

```
pwd
```

- ✓ Full path display

6 mkdir

 New folder create karta hai

```
mkdir test
```

7 touch

📁 New file create karta hai

```
touch file.txt
```

8 cat

📁 File content read/open karta hai

```
cat test.txt
```

9 rm

📁 File delete karta hai

```
rm test.txt
```

10 rmdir

📁 Empty folder delete karta hai

```
rmdir foldername
```

File Move Command

1 1 mv

📁 File ko move ya rename karta hai

```
mv test.txt /home/ravi/Documents
```

- ✓ File transfer
- ✓ Rename bhi possible

User Info Commands

1 2 whoami

👉 Current user ka naam show karta hai

```
whoami
```

1 **3** exit

👉 Terminal close / logout

```
exit
```

🔑 Terminal Control

1 **4** clear

👉 Screen clear karta hai

```
clear
```

⬆️ ⬇️ Arrow Keys

✓ Up arrow → previous commands

✓ Down arrow → next commands

👉 Repeated typing se bachata hai

🔍 Text Zoom

✓ Ctrl +

👉 Text size increase (terminal/browser dependent)

⚙️ System Update Commands (Linux)

1 **5** apt update

👉 Package list update karta hai

```
sudo apt update
```

1 **6** apt upgrade

apt upgrade

📁 Installed tools update karta hai

```
sudo apt upgrade
```

File Color Meaning (Linux)

Type	Color
File	White ○
Folder	Blue ●

Quick Command Summary

Command	Work
ls	list files
cd	change directory
pwd	current path
mkdir	create folder
touch	create file
rm	delete file
rmdir	delete folder
cat	open file
mv	move/rename
whoami	current user
sudo su	root access
clear	clear screen
exit	logout

GUI vs CLI Comparison

Feature	GUI 🖱️	CLI 🖥️
Ease	Easy	Hard
Speed	Slow	Fast
Control	Limited	Full



Interview Questions

- ✓ GUI aur CLI me difference kya hai?
- ✓ Linux me file delete ka command?
- ✓ sudo su kya karta hai?
- ✓ ls vs pwd difference?
- ✓ ifconfig ka use?



Pro Tip (Cyber + OS Insight)

👉 Hackers aur sysadmins mostly CLI use karte hain because:

- Fast control
- Remote access (SSH)
- Automation scripting



Final Summary

OS ka core idea simple hai:

- ✓ GUI = Easy but limited
- ✓ CLI = Powerful but technical

👉 Agar CLI strong ho gaya na,
to system control level next level ho jata hai 🚀